A Market World Model for Public LLMs: A Typed Observation Contract before Decision

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Abstract

Public LLMs asked to analyze or trade crypto markets fail before strategy: they lack a complete, honest, auditable *market observation interface.* We present an anonymized *world-model runtime* that compiles asynchronous multi-band evidence into a point-in-time (PIT) world bundle with a typed abstention contract. Epistemic observations $O_{j,t} = (x, \tau, q, g, r)$ and compilation Π_t yield $\mathcal{F}_t^{\mathrm{AI}}$; completeness, honesty (B, U, H), and WMI/ACWMI expose machine-readable refusal; evidence IDs make actions auditable. The primary claim is *trustworthy interfacing*, not a new trading agent: on identical 100 OOS asset-days, four live public LLMs (temperature 0) form a behavior ladder—asked *blind*, every model refuses (1.0); given a momentum fragment they over-trade (abstain 0.04–0.75) and lose; given the compiled world with a hard flag they abstain on every thin day (1.0); Ungated (same content, no hard flag) averages only 0.68. A full-OOS checkpointed sweep (n=2000) keeps Compiled at 1.000 for all vendors. Even on archive-thick days (all three evaluation bands ready), Raw still over-acts and often loses while Compiled continues to refuse—readiness counts alone do not replace the typed contract. A grounding workflow recovers ready/missing/tilt answers at mean F1/acc. 0.97 from Compiled vs. 0.04–0.23 from Raw. As a secondary *content* probe (not an LLM leaderboard), a transparent world-model rule that reads the same vintaged tilts beats momentum and buy-and-hold out-of-sample (Sharpe 0.76 vs. 0.10 / −1.40; ΔCE=0.334, p=0.034). We detail the runtime (collectors, vintage store, BandPIT, bundle schema, LLM adapter) as an ICAIF systems contribution.

CCS Concepts

• **Computing methodologies** → **Artificial intelligence**; *Machine learning*; • **Applied computing** → *Economics.*

Keywords

world models, public LLMs, AI agents, cryptocurrency, point-in-time, abstention, trustworthy AI, observation contract, systems

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1 Introduction

Public LLMs are increasingly asked to analyze or trade crypto markets. The bottleneck is not another alpha signal. It is *understanding*: before decision, an agent needs a market world that is **complete** (missingness disclosed), **honest** (stale or role-incoherent evidence gated), and **auditable** (actions bound to evidence). Venue fragmentation, derivatives, on-chain flows, news, and macro vintages arrive on incompatible clocks [13, 15]. Generative world models [7, 8, 11] and tool-using finance agents [22–24] typically assume a usable observation stream. Crypto agents often do not have one.

Thesis. We build a market *world-model runtime* for public LLMs: compile raw multi-band evidence into a PIT-safe bundle that GPT / DeepSeek / GLM / Gemini-class models can call under a typed refuse contract. The product objective is *trustworthy observation*, not a new portfolio policy. Strategy metrics appear only as a secondary probe that the compiled fields are nonempty. In slogan form: *quant from strategy; this runtime from understanding—enforced at the interface.*

What we claim. RQ1 tests whether a typed world interface can *enforce* world-conditional abstention across live public LLMs—a runtime contract, not a claim that models spontaneously discover thinness from raw ticks. RQ2 tests whether the compiled world makes analyst-facing epistemic answers verifiable. RQ3 is a *content* probe: when a transparent rule reads the same vintaged tilts, does the compiled world beat traditional momentum and buy-and-hold? We do *not* claim an LLM trading leaderboard.

Contributions.

(1) **Cognition semantics.** Epistemic observations, compilation Π_t, a lag reconstruction bound, completeness/honesty fields, WMI/ACWMI abstention, evidence-bound actions.
(2) **Anonymized runtime architecture.** Vintage-aware collectors, BandPIT previous-close clock, quality-tagged bundles, availability shocks O_t, OpenAI-compatible multi-vendor adapter, and a frozen Compiled / Ungated / Raw / Blind consumer protocol.
(3) **Live interface validation.** Four-arm ladder on identical days: Blind refuses, Raw over-trades and loses, Ungated is vendor-dependent, Compiled enforces thin-world abstain 1.0 (replicated on full OOS for all vendors). A band-thick split shows the contract still binds when all archive bands are ready; grounding recovers ready/missing/tilt at 0.97 vs. 0.04–0.23 raw; a secondary content rule confirms the tilts are economically nonempty.

2 Related Work

World models in AI.. World models learn compact states and often dynamics [7, 8, 11]. We address the complementary *observation-side* problem: a PIT-safe, quality-tagged crypto world *before* dynamics or LLM policy. We claim the missing systems layer generative WMs usually assume—not a Dreamer/JEPA simulator.

LLM agents in finance. ChatGPT return prediction [14], open financial LLMs [21], memory-augmented trading agents [12, 23], and tool-augmented agents [22, 24] advance agent skill and orchestration. Recent ICAIF work intensifies complementary layers of

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that stack: FinAgentBench evaluates agentic retrieval for financial QA [1]; FinSearch builds temporal-aware search agents over market and news feeds [18]; FinResearchBench judges long-horizon research agents with logic-tree protocols [19]; FactorMAD uses multi-agent debate for interpretable alpha mining [3]. Those lines improve *how* agents fetch, debate, and are scored. We address a prior gap they still assume—a PIT-safe, quality-tagged *observation contract* with typed abstention—and deliberately do not propose another memory router, retrieval planner, or debate society. Our estimand is within-model Compiled / Ungated / Raw / Blind under a frozen schema.

Selective prediction and trustworthy AI.. Abstention can be optimal under noise [4, 5]. We tie refusal to world quality via an explicit runtime contract, not classifier confidence alone.

PIT measurement and crypto microstructure. Macro vintages [2] and crypto microstructure [13, 15] motivate multi-band compilation. Bootstrap discipline for the economic probes follows [9, 17, 20]. Asset-pricing ML [6, 10, 16] is adjacent: we validate the economic content of a compiled world for agents, not priced factors.

3 Market World-Model Semantics

Definition 3.1 (Epistemic observation). An epistemic observation is

$$O_{j,t} = (x_{j,t},\ \tau_{j,t},\ q_{j,t},\ g_{j,t},\ r_{j,t}) :$$

value, latest-available time, quality, main-view gate, semantic role. A bare feature dump that omits (τ, q, g, r) is not a world observation.

Definition 3.2 (Financial world state). $W_t = \Pi_t(\mathcal{F}_t^{\text{raw}})$ with scalar quality $q(W_t)$ (e.g. WMI/ACWMI) gating abstention; $\mathcal{F}_t^{\text{AI}} = \sigma(W_t)$.

Definition 3.3 (Compilation). $\Pi_t = B_t \circ M_t \circ A_t$: align clocks, apply honesty/missingness gates, assemble band views.

Completeness, honesty, quality. Ready/limited/missing counts disclose coverage. $\text{WMI}_t = B_t U_t H_t$; ACWMI supports regime-conditional gating. The Compiled contract exposes the abstain flag (`should_ai_abstain`) and `thin_world` as first-class fields.

Assumption 3.4 (Bounded lag and increments). Band j has max observation lag Δ_j; latent state increments satisfy $\|S_u - S_{u-1}\| \le \bar{\delta}$.

Proposition 3.5 (Lag reconstruction bound). *Under Assumption 1,*

$$\|\widetilde{S}_t - S_t\| \ \le\ C_{\text{del}}\bar{\delta} + C_{\text{noi}}\varepsilon_t + C_{\text{miss}}m_t.$$

Compilation cannot erase delay; it can refuse to pretend the bound is zero.

Intuition. Crypto agents that concatenate tool dumps silently absorb delay and missingness into the feature vector. Prop. 3.5 makes those terms explicit: a thicker raw payload can worsen the reconstruction if honesty collapses.

Proposition 3.6 (Compilation ≠ feature dump). *Enlarging raw span without Π_t need not enlarge usable $\mathcal{F}^{\text{AI}}$: ungated stale evidence can raise volume while lowering honesty/stability.*

Proposition 3.7 (World-conditional abstention). *If non-abstain actions exceed a world-dependent cost $c_{\text{abs}}(W_t)$, the Bayes action is abstain; with costs decreasing in WMI, abstention forms a lower set in world quality [4].*

Table 1: Anonymized module map (systems view).

Module	Responsibility
Collectors	Ingest multi-band series with timestamps / vintages
Vintage stores	Persist raw, market, and analytics objects
BandPIT compiler	Previous-close readiness, Π_t, WMI/ACWMI
Bundle builder	Assemble complete/honest/auditable JSON world state
Shock index O_t	Query availability outages for quasi-exogenous levers
LLM adapter	OpenAI-compatible chat; frozen prompts; temp. 0
Eval harness	Compiled/Ungated/Raw/Blind sampling, transcripts, tables

Proposition 3.8 (LOBO content vs. gating). *Deleting band j changes value through a content channel and a gating channel. Empirically we report content share under an ungated rule so the probe speaks to nonempty fields.*

Reading for systems reviewers. Props. 3.6–3.7 justify why the runtime exports gates and abstain flags rather than only tilts. Prop. 3.8 justifies the economic probe: if deleting a durable band does not change actions, the bundle is decorative—and the probe confirms it is not.

Remark 3.9 (Not a generative WM). *We do not learn $p(s_{t+1} \mid s_t, a_t)$. We deliver a* state compiler + abstention runtime.

4 Runtime Architecture

We describe the anonymized prototype as a layered systems object (Figure 1; module map in Table 1). Source paths and product names are omitted for double-blind review; an anonymized artifact will be released upon acceptance.

4.1 Layered design

(1) **Data layer (collectors).** Exchange, macro, and alternative collectors write vintage-aware history: observation timestamps plus macro `available_at` vintages.
(2) **Store layer.** Embedded analytical stores hold raw series, merged market panels, and pipeline outputs (readiness, WMI/ACWMI, paper world snapshots).
(3) **Logic layer (compilation).** BandPIT reconstructs band readiness on a previous-close clock; honesty/missingness gates and band assembly implement Π_t; world-quality indices and availability shocks O_t are first-class.
(4) **Service layer.** A read API serves quality-tagged AI bundles and health/readiness objects to consumers without requiring collector restarts for newly arrived rows.
(5) **Consumer layer.** Frozen-prompt LLM / rule agents call an OpenAI-compatible adapter (temperature 0) under Compiled, Ungated, or Raw treatments.

4.2 Data flow: inputs, processing, outputs

Table 2 traces the runtime end-to-end. *Inputs*: the evaluation archive populates three public bands—exchange klines/funding/open-interest

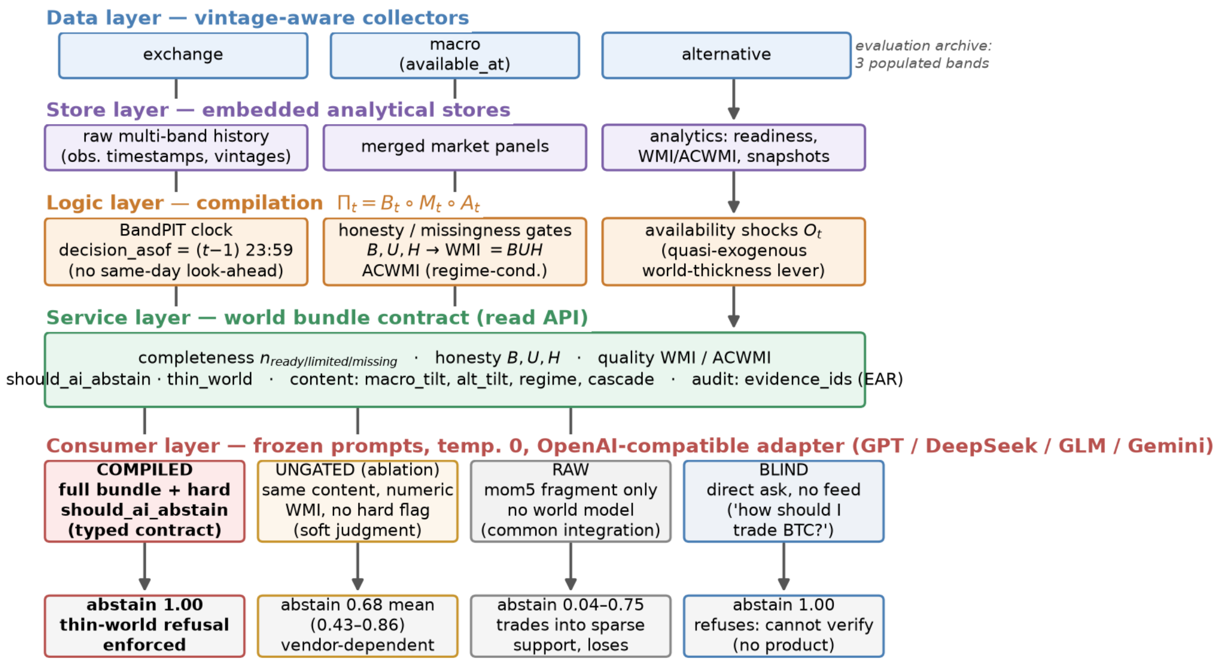

Figure 1: Anonymized runtime architecture. Five layers map raw multi-band evidence to a typed world-bundle contract; the consumer layer runs the frozen four-arm protocol (Compiled / Ungated / Raw / Blind), whose live abstention outcomes are shown at the bottom.

via REST, daily macro risk indices (equity volatility, dollar index) with `available_at` vintages, and aggregate stablecoin circulation—each row stored with its observation timestamp. *Processing*: for every asset-day, BandPIT selects the latest observation with timestamp ≤ (t−1) 23:59, grades it fresh/stale/missing against per-band age thresholds, aggregates breadth/stability/honesty into WMI and regime-conditional ACWMI, derives content tilts (5-day macro-index changes; 7-day stablecoin-flow slope; return momentum), and logs availability shocks O_t. *Outputs*: a ∼3,990-row PIT panel (10 assets × 399 days), per-day world bundles scoped to these three archive bands (Listing 1), and the consumer transcripts/tables behind Section 5.

4.3 PIT previous-close clock

For calendar day t, the decision information set is reconstructed at (t−1) 23:59. The payoff is the same-calendar-day close-to-close return r_t. This removes same-day look-ahead in band statuses and vintaged macro/alternative content. The bundle records the clock in `decision_asof`, and every live run pins `timing_protocol` to the previous-close protocol.

4.4 World bundle contract

Table 3 lists field groups. Listing 1 shows a compact Compiled example from the OOS panel: exchange missing so 2/3 archive bands ready, WMI low, `should_ai_abstain=true`, and `evidence_ids` binding actions to disclosed bands/tilts. Ungated removes the hard

Table 2: Data flow through the runtime (three-band evaluation archive).

Band	Fetched (raw)	Emitted (bundle)
exchange	OHLCV klines (1h/1d), funding, open interest, orderbook	mom5; band status/age; funding context
macro	equity-vol and dollar indices; vintaged macro series	macro_tilt (5d changes); status/age
alternative	aggregate stablecoin circulation	alt_tilt (7d flow slope); status/age
payoff	daily closes (external reference)	close-to-close r_t for scoring only
derived	—	B, U, H, WMI, ACWMI, `should_ai_abstain`, `thin_world`, O_t, `evidence_ids`

boolean (and threshold) but keeps numeric WMI/ACWMI and completeness. Raw retains only `mom5` (plus a non-informative noise bit).

Listing 1: Compact Compiled bundle example (OOS exchange gap).

```
{"date":"2026-05-11","asset":"ADA",
 "decision_asof":"2026-05-10T23:59:00",
 "completeness":{"n_ready":2,"n_missing":1,
```

Table 3: World bundle field groups (cognition base).

Group	Fields
Timing	`decision_asof`, previous-close protocol
Completeness	$n_{\text{ready/limited/missing}}$, band statuses
Honesty	B, U, H, main-view / stale gates
Quality	WMI, ACWMI, `should_ai_abstain`, `thin_world`
Content	macro_tilt, alt_tilt, regime, cascade
Audit	`evidence_ids`, EAR required

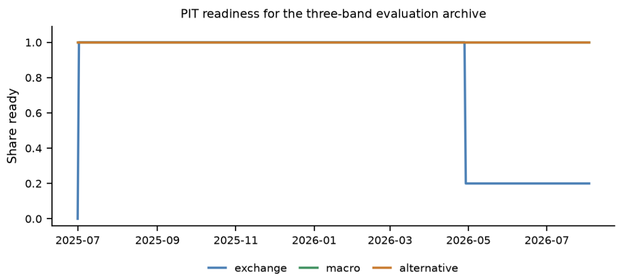

Figure 2: PIT readiness for the three-band evaluation archive.


```
  "ready_share":0.67,"archive_bands":
  ["exchange","macro","alternative"]},
"world_model_index":{"wmi":0.015,"acwmi":0.30,
  "should_ai_abstain":true,"thin_world":true},
"macro_tilt":1.0,"alt_tilt":-1.0,
"detected_regime":"crisis",
"audit":{"evidence_ids":[
  "band:exchange:missing","band:macro:ready",
  "band:alternative:ready","tilt:macro:1.0"]}}
```


4.5 Four information arms and LLM adapter

- **Compiled (product contract):** full bundle; frozen prompt *requires* abstain when `should_ai_abstain` is true.
- **Ungated (strong content control):** same content/completeness/numeric WMI; *no* hard flag; soft prompt asks the model to judge thinness. This is the non-strawman ablation: disclosure without enforcement.
- **Raw (common thin integration):** `mom5` only; no quality index or abstention guidance—matching the widespread "price signal in, action out" wiring, not a cartoon baseline.
- **Blind (direct ask):** the retail question verbatim—"Today is d. How should I trade a?"—with *no* data feed at all.

Actions $\in$ {bullish, bearish, neutral, abstain}. Primary metrics are thin-world abstain rate and EAR proxy; CE is reported as a safety side-effect (avoided loss), not as an alpha claim. The evaluation archive is the three-band set {exchange, macro, alternative} (Figure 2): macro and alternative stay ready throughout; exchange readiness drops in the later OOS window and defines the within-archive *band-thick* contrast below. In this scarce panel WMI stays below the refuse threshold on every asset-day, so Compiled abstention is the intended stress test of the contract under sparse support. The adapter speaks OpenAI-compatible chat completions; credentials stay in environment variables (never in artifacts).

4.6 Frozen prompt contracts

Listings 2–3 excerpt the frozen Compiled and Ungated contracts (temperature 0; action schema identical). The only intentional difference is hard vs. soft abstention. Raw prompts omit world-quality language entirely.

Listing 2: Compiled contract (excerpt).

```
Decision rules (excerpt):
1. If world_model_index.should_ai_abstain is true
   OR the world is thin, you MUST "abstain".
2. Prefer band content over raw momentum
   (macro_tilt, alt_tilt are PIT-safe).
3. Rationale must be consistent with evidence_ids.
```

Listing 3: Ungated soft contract (excerpt).

```
Decision guidance (soft):
1. Judge whether missingness / low WMI/ACWMI
   makes action unsafe; you MAY "abstain".
2. Prefer band content when you act.
NO hard abstain boolean; NO forced policy.
```

Not "just instruction following." Compiled abstention is partly prompt obedience—and that is the *product requirement*: production agents need an enforceable refuse interface when $q(W_t)$ is low (Prop. 3.7). Ungated measures soft judgment from disclosure alone; Raw measures the common "price signal only" integration. Reliable Compiled refusal is the SLA; spontaneous Ungated refusal is a bonus.

4.7 Service surface and reproducibility

The service layer exposes read-only objects used by the consumer harness: (i) quality-tagged world bundles; (ii) band readiness / WMI–ACWMI snapshots; (iii) availability-shock queries O_t; (iv) health metadata disclosing whether paper engines or production proxies hydrate ACWMI inputs. Collectors and the logic pipeline can run as supervised jobs; the API reads stores live, so newly compiled rows appear without process bounce. For review we freeze prompts, temperature, action schema, IS/OOS cut, and sampling seed; model IDs and an OpenAI-compatible base URL are configuration, not estimands. Credentials never enter committed artifacts. Upon acceptance we will release an anonymized repository with the consumer harness, frozen prompts, and table-reproduction scripts.

5 Experiments

5.1 Setup

PIT panel ~399 days, 10 liquid names, previous-close clock, chronological IS/OOS 200/200 days (≈2,000 OOS asset-days). Live RQ1 uses the *same* stratified 100 OOS asset-days across arms (seed fixed), temperature 0, frozen prompts, and an OpenAI-compatible gateway; a checkpointed full-OOS sweep tests population scale. Vendor models are GPT-, DeepSeek-, GLM-, and Gemini-class flash/mini IDs listed in Table 4. Scarce/thin labels bind on OOS by design; Sec. 5.4 covers fully populated days.

Table 4: Live abstention ladder on identical 100 OOS asset-days (Wilson 95% CIs); Blind abstains 1.0 for every model. ΔCE is avoided-loss, not an alpha claim.

Model	Thin C	Thin U [CI]	Abs. R [CI]	ΔCE
gpt-5.4-mini	1.00	0.63 [.53,.72]	0.04 [.02,.10]	+0.73
deepseek-v4-flash	1.00	0.81 [.72,.87]	0.75 [.66,.82]	+1.22
glm-5.2	1.00	0.86 [.78,.91]	0.11 [.06,.19]	+1.04
gemini-3.5-flash-lite	1.00	0.43 [.34,.53]	0.07 [.03,.14]	+1.30
Mean	1.00	0.68	–	+1.07

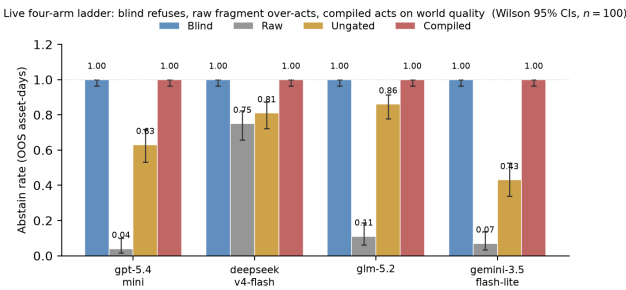

Figure 3: Live four-arm interface ladder (same 100 days; Wilson 95% CIs). Blind refuses; Raw over-trades; Ungated is vendor-dependent; Compiled enforces the typed refuse contract.

Table 5: Raw-arm action mix (100 days). Compiled abstains on all thin days; Blind (direct ask, no feed) abstains 100% for every model.

Model	Abs.	Bull.	Bear.	Neut.
gpt-5.4-mini	0.04	0.28	0.38	0.30
deepseek-v4-flash	0.75	0.14	0.04	0.07
glm-5.2	0.11	0.40	0.48	0.01
gemini-3.5-flash-lite	0.07	0.44	0.49	0.00

5.2 RQ1 (primary): Live interface behavior across information arms

Table 4 and Figure 3 are the headline result; all arms share the same 100 OOS asset-days. Under Compiled, all four models abstain on every thin-world day (thin-abs. 1.0); rationales cite world fields on 93–99% of days—this is prompt-contract obedience by design (the product SLA), not a claim of spontaneous thinness discovery. Under Ungated, mean thin-abs. falls to 0.68 (range 0.43–0.86): DeepSeek and GLM stay above 0.80, GPT drops to 0.63, and `gemini-3.5-flash-lite` falls to 0.43 and trades. When Ungated models do act on this 100-day draw, they lose (CE −1.04 to +0.02)—soft judgment is unstable exactly where enforcement matters. Under Raw, abstain rates are 0.04–0.75 with substantial directional mass (Table 5). Safety side-effect: mean ΔCE (Compiled−Raw) = +1.07, positive for every vendor, because Compiled sits out the losses Raw incurs on sparse support.

Table 6: Full-OOS band-thick split (3/3 ready; WMI still thin). Compiled refuses; Raw over-acts.

Model	Abs. U	Abs. R	CE R
gpt-5.4-mini	0.58	0.00	−0.28
deepseek-v4-flash	0.42	0.75	+0.04
glm-5.2 (n=398 R)	0.58	0.09	−0.87
gemini-3.5-flash-lite	0.34	0.00	−0.25

Full-OOS replication. To rule out subsample luck we checkpointed the three data arms on the *full* OOS panel (n=2000 asset-days; every OOS day is thin-world under the scarce-support design). Compiled thin-abstain is 1.000 [.998, 1.000] for *all four* vendors—the typed refuse contract is a population property, not a lucky draw. Ungated rates tighten the Wilson intervals and keep the soft-judgment ladder, with mean thin-abs. 0.56 (GPT 0.666 [.645, .686]; GLM 0.679 [.654, .703] on n=1402 completed calls; DeepSeek 0.540 [.518, .561]; Gemini 0.359 [.338, .380]). Vendor *ranking* within Ungated shifts relative to the 100-day sample (DeepSeek is less cautious at scale), but no vendor matches Compiled. Raw fragment over-trading remains decisive where the scrap is trusted: GPT and Gemini abstain at 0.000 [.000, .002]; DeepSeek stays cautious at 0.748 [.728, .767] (n=1925). Because the full OOS window is entirely thin, Compiled CE is identically zero by construction—thick-day economics are the dense-world probe of Sec. 5.4, not this honesty stress test.

The Blind arm: what happens if you just ask. The product-realistic control is the question a retail user actually types: *"Today is d. How should I trade BTC?"* with no data feed. Run live on the same 100 asset-days, all four models abstain on 100% of days (CE = 0; zero directional actions). A bare public LLM is not a market agent—it refuses. The failure mode that costs money is the middle case: give the same model a *fragment* (Raw) and it starts trading (abstain as low as 0.04) and loses. Compiled matches Blind's refusal on thin days *by contract*, while still exposing the structured world that RQ2 shows is analytically usable and that RQ3 shows is economically nonempty when consumed by a transparent rule.

Half-B patch: band-thick days still need the contract. Archive readiness alone is not a refuse policy. On full-OOS *band-thick* days (exchange+macro+alternative all ready; 61% of OOS, $n \approx 1224$), WMI still marks every day thin under the scarce-support design, so Compiled abstains at 1.0. Ungated soft judgment only partially recovers (mean abs. ≈0.48; Gemini 0.34). Raw remains the dangerous middle: GPT and Gemini abstain at 0.0 with CE −0.28 and −0.25. Table 6: when the archive *looks* complete, fragment feeds still over-act and lose, while the typed contract continues to refuse—exactly the production failure mode a readiness dashboard would miss.

Inside the Ungated arm. Ungated rationales cite disclosed world fields on 53–93% of days, so models *read* the world; they differ in acting on it. Vendor-specific caution—not missing disclosure—drives the gap the typed contract closes.

Offline diagnostics. World-honoring offline followers abstain at 1.0 on full OOS; momentum-only mocks ignore thin-world flags (thin-abs. 0). Offline results are protocol checks; live vendors are the product claim.

Table 7: Grounding workflow (50 days): ready / missing / tilt signs.

	Ready F1		Missing F1		Tilt-sign acc.	
Model	C	R	C	R	C	R
gpt-5.4-mini	0.94	0.00	0.94	0.31	0.94	0.20
deepseek-v4-flash	0.98	0.06	0.98	0.20	0.98	0.18
glm-5.2	0.98	0.09	1.00	0.11	0.98	0.20
gemini-3.5-flash-lite	0.96	0.03	0.96	0.30	0.96	0.18
Mean	0.97	0.04	0.97	0.23	0.97	0.19

5.3 RQ2: Non-trading cognition—grounding workflow

The cognition base should serve analysis, not only trade/abstain decisions. We run a live multi-step *grounding workflow* on 50 OOS asset-days per arm. Given Compiled or Raw inputs, the model must (i) list which of {exchange, macro, alternative} are ready, (ii) list which are missing, (iii) report sign(macro_tilt) and sign(alt_tilt), and (iv) judge sufficiency. Steps (i)–(iii) are scored against PIT ground truth. Table 7: with the Compiled bundle, mean ready-band F1 0.97, missing-band F1 0.97, and tilt-sign accuracy 0.97; with Raw, the same scores collapse to 0.04, 0.23, and 0.19. Sufficiency alone is *not* the discriminator once completeness is scoped to real archive bands—models may call a 3/3-ready day "sufficient" while WMI still marks the world thin—so the product claim is *verifiable epistemics*: only the compiled world lets the model say what is ready, what is missing, and which tilts are present, and be checked. This is the analyst-facing counterpart of Prop. 3.5's missingness term.

5.4 RQ3: Secondary content probe (not an LLM leaderboard)

RQ1–RQ2 establish the interface. RQ3 asks a narrower question: are the compiled tilts economically nonempty, or decorative? A transparent rule shares return inputs with momentum and adds vintaged macro_tilt / alt_tilt—no LLM in the loop. Out-of-sample (199 days $\times$ 10 assets) this content rule beats traditional baselines (Table 8, Figure 4): Sharpe 0.764 / CE +0.130 versus momentum (0.096, -0.206) and buy-and-hold (-1.404, -1.163). The pre-specified contrast mechanism − momentum is significant under block bootstrap (ΔCE=0.334, p=0.034). ΔCE is positive for all ten assets (Figure 4, right). LOBO (Prop. 3.8) attributes the gain to band content (deleting macro/alternative collapses to momentum); gaps survive 10–25bps costs. A 2017–2026 audit with non-vintaged proxy tilts shows no edge (ΔCE $\approx$ 0, p=0.81): value appears where the vintage-compiled world exists. This supports "the bundle is not empty," not "our LLM is a trading SOTA."

Within-archive dense world. Among the three archive bands, 61% of OOS asset-days are band-thick (3/3 ready) and 39% are band-thin (exchange gap; Figure 2). Table 9: the mechanism−momentum gap is stronger on band-thick days (ΔCE=0.316) than on exchange-gap days (0.129)—the content channel pays most when the archive is populated, complementary to RQ1's refuse-under-thin stress test.

Table 8: Secondary content probe: world-model rule vs. traditional baselines (no LLM).

Policy / contrast	Sharpe	CE
Buy-and-hold (always long)	−1.404	−1.163
Momentum always	0.096	−0.206
World-model content rule	**0.764**	**+0.130**
Mechanism − Momentum	—	0.334 (p=0.034)

Table 9: Dense-world split of the content probe (OOS; three archive bands).

Slice	Share	Mech CE	ΔCE vs mom
All OOS	1.00	0.130	0.336
Thick (3/3 ready)	0.61	0.059	0.316
Thin (exchange gap)	0.39	1.048	0.129

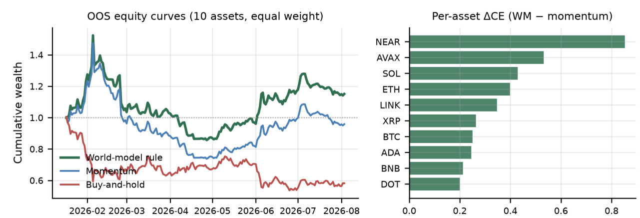

Figure 4: Secondary content probe (no LLM). Left: equity curves; right: ΔCE vs. momentum positive for every asset.

5.5 Discussion and threats

The product evidence is the interface ladder: Blind refuses (no product), Raw over-acts and loses, Ungated is vendor-dependent ($\approx$0.68; GLM 0.86 vs. Gemini 0.43), Compiled enforces thin-world abstain 1.0—including on band-thick days where a readiness dashboard would look green. Claim: *understanding first, enforced at the contract; content economics are secondary.*

Relative to FinMem / FinAgent-style systems [23, 24] and ICAIF retrieval/debate/judge lines [1, 3, 18, 19], we do not propose a new memory, search, or debate router. We propose the observation compiler and refuse SLA those stacks need when feeds are asynchronous and quality-heterogeneous. Relative to return-prediction LLM studies [14], we score refusal and auditability before PnL.

Design choices and validity. (1) *Prompt contract*—Compiled refusal is the product SLA; Ungated is the strong same-content control. (2) *Three-band archive*—every readiness label is verifiable; band-thick vs. band-thin separates archive coverage from WMI thinness. (3) *Baselines*—Raw matches common mom-only wiring; RQ3 baselines are momentum/buy-and-hold for the content probe only. (4) *Sampling*—100-day headline ladder plus full-OOS checkpoints (n=2000; GLM Raw incomplete). (5) *What we do not claim*—LLM trading SOTA, generative dynamics, or spontaneous thinness discovery without the boolean.

6 Design Rationale for ICAIF Systems Readers

Three implementation choices are deliberate.

Export gates, not only features. Silent feature stores hide staleness. By placing completeness, honesty, and `should_ai_abstain` in the bundle, the runtime makes refusal a first-class machine-readable event rather than a free-text apology.

Previous-close PIT over wall-clock streaming. Intraday streaming demos look impressive but confound look-ahead. The previous-close clock is conservative and auditable: every live decision cites a `decision_asof` stamp that a reviewer can recompute from the panel.

Four arms instead of one leaderboard. A single Compiled-vs-holdout Sharpe invites Holy-Grail readings. Compiled / Ungated / Raw / Blind separates (i) typed enforcement, (ii) soft judgment from disclosure, (iii) common thin integrations, and (iv) no feed—matching how people actually query LLMs about markets, without turning the paper into a trading contest.

7 Scope and Future Work

This study delivers an observation-side world-model *interface*: a state compiler with typed abstention on a three-band public archive. We do not claim LLM portfolio SOTA, generative dynamics, or coverage of news/on-chain/options until those vintages exist. Natural extensions: additional evidence domains, learned dynamics on the compiled state, richer analyst workflows, and optional thick-regime live trading studies once WMI leaves the scarce floor. Camera-ready will restore author identity and release an anonymized harness (frozen prompts, sampling seed, table-reproduction scripts) deferred during review.

8 Conclusion

Public LLMs need a market they can *safely observe* before they decide. We contribute cognition semantics and an anonymized layered runtime that turns asynchronous crypto evidence into a complete, honest, auditable world bundle with a typed refuse contract. Live public LLMs exhibit a four-arm interface ladder—Blind refuses, Raw over-trades and loses, Ungated is vendor-dependent, Compiled enforces thin-world abstain 1.0 even on band-thick days—and the same bundle makes analyst answers verifiable (grounding 0.97 vs. 0.04–0.23). A secondary content rule shows the tilts are economically nonempty versus momentum and buy-and-hold. Understanding comes first—enforced at the interface.

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